

Chapter – 7: Temperature and Its Measurement

Wrong measurements are worse than no measurements at all. – Anna Mani

- Labok and his sister, Phiban – live in Shillong
- One day – return from school – parents were out
- Labok complaint – feverish – sister – checked forehead – might be true
- She took out thermometer – washed with soap, water – took Labok's temp.
- Luckily – temp. was normal – washed thermometer again – dried up – placed in almirah again
- Asked Labok – change uniform, eat lunch, rest some time

Hot or Cold

- Some bodies – hotter than others
- Ex. – summers – tap water – hotter than – *matka* (earthen pot) or refrigerator
- Realise this – mere touch – BUT – not always reliable
- Activity –
 - Take – 3 large containers – A, B, C
 - Container A – warm water – container B – tap water – container C – ice-cold water
 - 1st **predict** – what you feel
 - Dip your right hand – A – left hand – C – keep there – 2-3 minutes
 - Take hands out – place in B
 - Write predictions – what you feel
 - Right hand – dip in B
 - Left hand – dip in B
 - Now conduct act. – write obs.
- **Compare** – obs. match with predictions
- Right hand – container B – cool
- Left hand – container B – same water – warm
- These obs. – **inference** – cannot rely – sense of touch

Temperature

- Reliable measure – hotness (coldness) – **temperature**
- Hotter body – higher temperature than – colder body
- Diff. in temp. – tell us – how hot – a body – compared to – another body
- Device – measure temp. – **thermometer**
- 2 kinds of thermometers –
 - Clinical – Measure human body temp.
 - Laboratory – Measure temp. in labs, chem. exp.

Measuring Temperature

Clinical thermometer

- Familiar with thermometer – shown here – measure body temp.
- Such thermometers – display – digital – run on batteries



- Measure temp. – place in contact – person's body
- Measure temp. – thermometers use scale – **Celsius scale**
- This scale – unit of temp. – **degree Celsius** – denoted as – °C
- Earlier – mercury thermometers – used – BUT – mercury – very toxic, difficult to dispose
- Digital thermometers – use heat sensors – measure temp.
- Activity –
 - Wash – hands, tip of thermometer – soap, water
 - Reset thermometer – press reset button
 - Place it – under tongue – close mouth
 - Wait – thermometer – beeps or flashes light
 - Take out of mouth – read temp. – **record**
 - Clean tip – soap, water – dry it
 - Repeat above steps – measure temp. – your friends
- Normal temp. – human body – 37.0 °C
- Temp. – every person – diff. – may be higher OR lower than – normal temp.
- Normal temp. – avg. – large no. – healthy person
- Body temp. – influenced – several factors – age, time of day, activity level
- Try measuring temp. – diff. times, diff. days – record for a month – analyse
- Humans – temp. range – 35 °C to 42 °C
- Small children – old people – measure temp. – armpits – 0.5 °C to 1 °C lower than actual
- Another scale – **Fahrenheit scale** – unit of temp. – **degree Fahrenheit** – denoted as – °F
- Temp. on Celsius scale – 37.0 °C – same temp. on Fahrenheit scale – 98.6 °C
- Fahrenheit scale – not used anymore – scientific studies
- Scientific work – another scale – **Kelvin scale** – unit – **Kelvin** – denoted as – K – **SI unit**
- All 3 scales – named after scientists – developed them

Laboratory thermometer

- Long, narrow, uniform glass tube – sealed
- One end – bulb – contain liquid – outside bulb – narrow column of liquid
- Celsius scale – marked along tube
- Liquid column – rise or fall – temp. variations
- Mark of Celsius scale – coincide with top – liquid column – temp. reading
- Activity –
 - Take a lab. thermometer – obs.
 - Note down – lowest temp., highest temp.
 - Calculate range
- Range of lab. thermometer – -10 °C to 110 °C
- Activity –
 - Take – lab. thermometer – obs.
 - Note down –
 - Temp. diff. – 2 bigger marks
 - No. of divisions between 2 bigger marks
 - Calculate – temp. – smallest division – read
- Temp. diff. – between 2 bigger marks – 0 °C and 10 °C – OR – 10 °C and 20 °C – same – 10 °C
- No. of divisions – these marks – 10 – one small division – $10/10 = 1$ °C
- Smallest value – this thermometer – read – 1 °C

- School lab – thermometers – diff. range, diff. smallest value – look carefully – before start

Correct way of measuring temp. using a laboratory thermometer

- Thermometer – immersed in water – bulb – do not touch – bottom or walls – beaker
- Thermometer – hold vertically – not tilted
- Temp. – read while in water
- While reading – eye – in line – level of liquid column
- Activity –
 - Take – warm water – beaker
 - Dip – thermometer – bulb – immersed in water
 - Obs. – rise – liquid column
- Wait – column stops rising – measure temp. – do not wait too long – water cools down
- Take the thermometer out – liquid column – falls down
- Read temp. – thermometer in water
- Activity –
 - Phiban's teacher – arranged setup – measure temp. – boiling water
 - Phiban and classmates – measure temp. – note down
 - Compare – reading – diff. students
- Diff. in readings – **discuss**

Air temperature

- Some thermometers – hung on wall – school labs, doctor's clinic, hospitals, etc.
- These meters – approx. idea – room temp.
- Weather reports – mention – max., min. air temp. of the day
- Activity –
 - Read, listen – weather report – same place – 10 days
 - Record – max., min. air temp.
 - Analyse the data
- Max., min. temp. – changes everyday
- Weather – depend on – several factors
- Summers – temp. rises – winters – temp. falls
- **Anna Mani** – Indian scientist – weather woman – invented – large no. – weather measurement instruments
- Also explored – possibilities – wind, solar energy – helped India – global leader – renewable energy